Bionic Eye

Moorfields Eye Hospital has managed to successfully perform the world's first bionic eye transplant. An 88-year-old woman was the first recipient of the bionic eye implant, which has now allowed her to continue to pursue her favourite hobbies again, including gardening and watercolour painting. After seeing positive results, it looks like a lot more people stand to benefit from the Europe-wide clinical trial, which is being backed by the National Institute of Health Research (NIHR) Biomedical Research Centre at Moorfields and the UCL Institute of Ophthalmology.

If you're wondering how the bionic eye even works, surgeons implanted a 2mm-wide microchip behind the elderly woman's left eye, where she was blind. Then, they provided her with special video glasses which could capture the scene in front of her, and relay that information to the chip. This chip then relayed the same information to her brain, imitating normal vision. This transplant could potentially be a great solution for the problem of decaying eyesight with age and more.

This operation will be used to learn more about the surgery, any possible repercussions, and collect evidence, in general, to enable more people to get bionic eye transplants in the future.

Sony NW-WM1ZM2

Sony has just come out with the NW-WM1ZM2 Walkman, which is a successor to the NW-WM1Z which came out in 2016. This is a gold-plated Walkman from Sony. Yes, you read that right. Naturally, it costs as much as you'd expect something gold-plated to cost, at a whopping $4500. The NW-WM1ZM2 looks like a premium gadget, and if comparisons had to be drawn, similar to the thicker smartphones from back in the day. The chassis is made with high-grade gold-plated oxygen-free copper, which plays a role in providing the Walkman's "unique natural, acoustic sound", according to Sony. What Sony's promising here is a supposedly unrivalled audio performance, in a portable music player. We've got a 5-inch touchscreen on the front, and audio control buttons on the right side of the device as well. For connectivity and charging you've got a USB Type-C port, and a memory card slot as well, allowing you to further expand its 256GB of in-built memory. The device can play most lossless music formats, and comes with Sony's LDAC Bluetooth codec. According to Sony, the NW-WM1ZM2 can run for 40 hours on a single charge.

Personalised spinal cord implant

Dr. Gregoire Courtine and his colleagues at the Swiss Federal Institute of Lausanne (EPFL) have successfully managed to help a paralysed person walk with their invented spinal cord implant. Furthermore, it took Michel Roccati, the paralysed person in question, just one day of stimulation to get started. The implant is the first of its kind, which works by mimicking the electric signals from the brain to control lower body movements. Electrodes are placed directly onto the spinal cord, which gives access to the various neurons that control the lower body. Then, with the help of a tablet, the implants can be adjusted to support different activities, such as standing, walking, or even swimming and biking. With the help of the implants, Michel Roccati, and others who were part of the same trial, were able to traverse the streets of Lausanne with the aid of a walker. Spinal cord injuries usually sever the connection between the brain and the affected muscle groups. However, the affected muscle groups, despite no longer receiving signals from the brain, are still generally in working condition. The implant recreates the electrical signals that the brain would otherwise send in order to get them working again.